

The Forge

A FIELD CODEX ·
BALANCE SANDBOX

DESIGN SAMPLE · SEED

Game Systems & Mechanics

Designer

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This is a small, playable proof for one piece of The Forge. Think of it as a workshop bench, not a finished game. On it, a single system (who controls which ground, and how) can be tuned and tested across thousands of trials, long before a real crowd ever gathers. That is what a **balance sandbox** is: a place to learn what is fair and what is broken while it is still cheap to fix. The role asks for a living design document, balance logic, and playtest frameworks rather than finished code. That is exactly what this is.

PLATE I · FIELD SURVEY

LAKE SAN ANTONIO · SRTM DEM

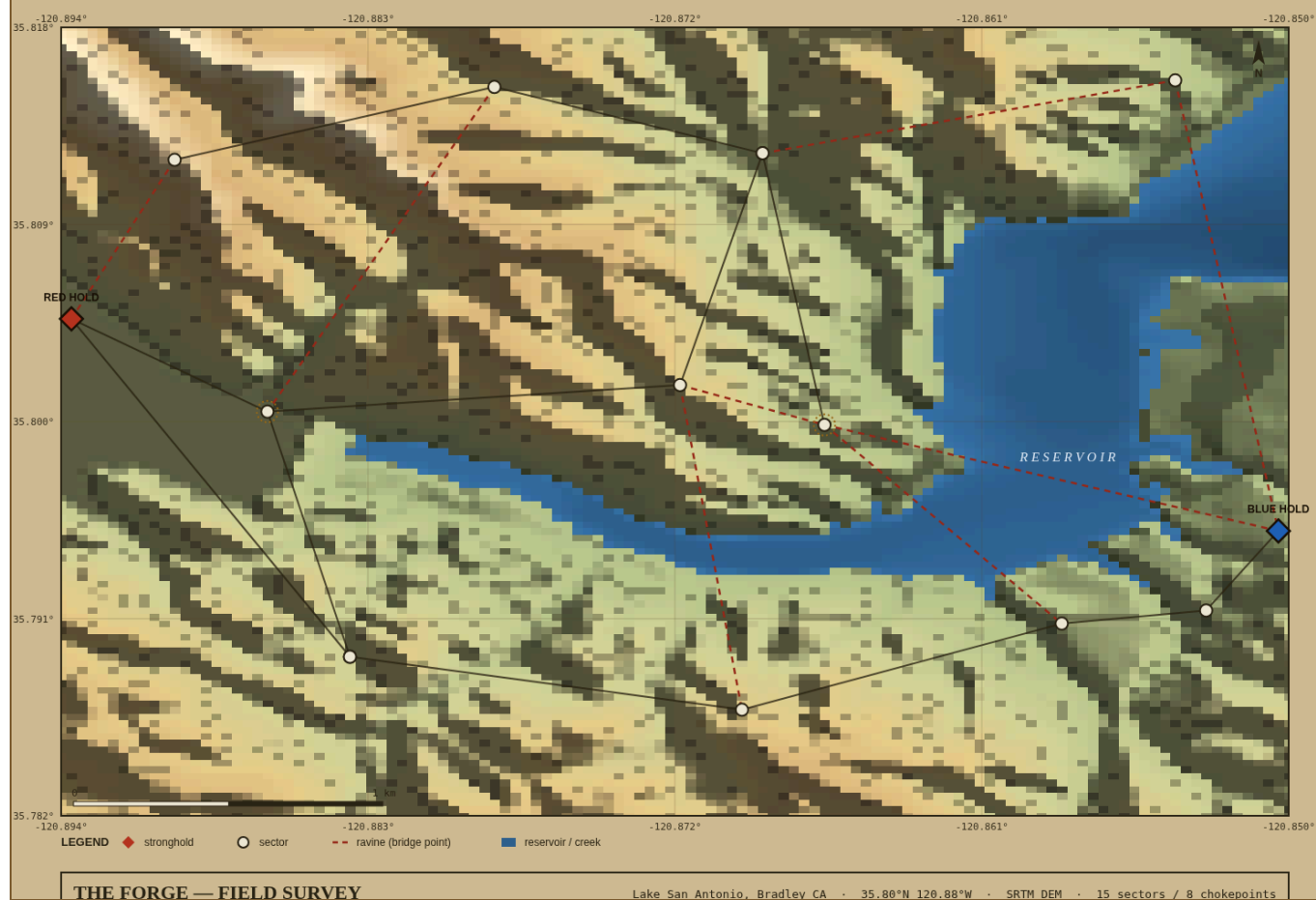


Plate I. A survey of the real ground. Blue is the reservoir and its creeks, the ravine a bridge must span. The red and blue diamonds are the home strongholds. Dashed red lines mark the crossings where a bridge may be raised. Contours, scale, and bearing are read straight from the elevation data.

I. THE FIELD

Every match is fought on a map, and the one above is real. It is built from a **DEM** (a digital elevation model, which is simply a height map of actual ground) of Lake San Antonio in California, a site that has hosted the Lightning in a Bottle festival. The reservoir and its creeks become the **ravine**, a barrier no one can cross on foot. The only way over is a **bridge**, and bridges must be built and paid for. The two diamonds are **strongholds**, the home keeps each side defends. Travel time follows the real slopes, so a climb costs more than a stroll. The point is plain. The terrain itself decides where the chokepoints are. Nobody drew them by hand.

II. How A MATCH Is Won

A side wins by holding ground over time. You take a sector by standing on it with more bodies than the enemy, which slowly turns it to your colour. You keep it by staying. Leave, and it drifts back to neutral, so no one can sit on the whole map at once. Players come in **classes**, each with a trade: scouts who see far, anchors who hold hard, engineers who raise bridges, quartermasters who enrich the land they stand on, heralds who broker deals. A match runs ninety minutes in four acts, from the opening land grab to a closing siege, paced by a costumed Game Master. The complete rules, with every class and number, live in the companion **Field Manual**.

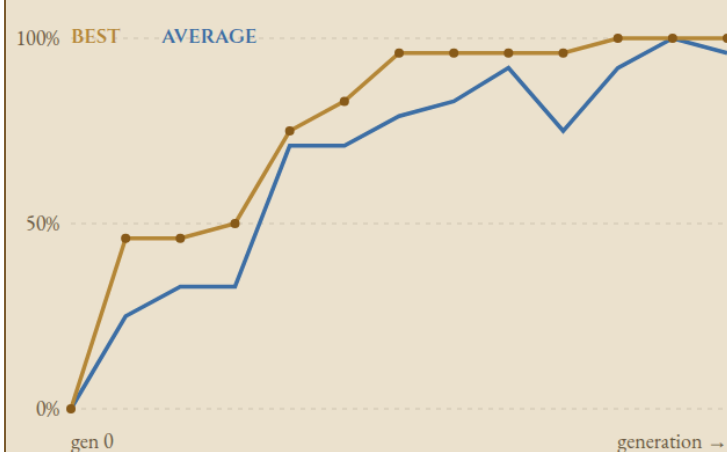
III. THE TIRELESS PLAYTESTER

To test all of this without gathering a crowd, the bench plays against itself. **Self-play** means the computer fields simple robot captains, lets a learner compete against them, and keeps whatever wins, round after round. The chart below is that learning happening. It begins at zero, a blank mind that plays badly, and climbs until it beats every hand-written strategy. The bars beside it show what the winner came to value. Read together they tell a story: it taught itself to rush the late-game siege and to stop wasting time guarding ground it already held. The lesson for a designer is blunt. A learning opponent will find the exploit your scripted bots never could. This is the honest use of such an AI here. It is a tireless playtester, not a spectacle.

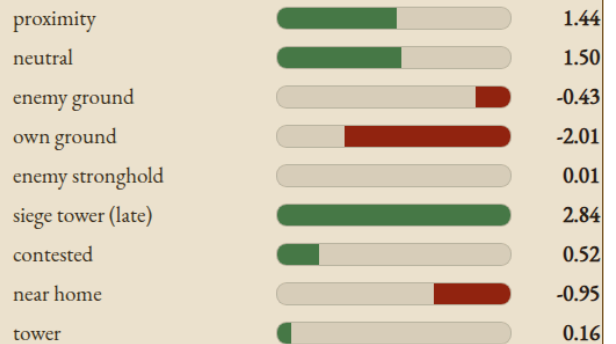
SELF-PLAY LEARNING CURVE

A trained playtester learns *The Forge* from nothing, by playing it over and over.

WIN-RATE AGAINST THE FOUR SCRIPTED STRATEGIES



WHAT THE WINNING POLICY LEARNED TO VALUE



Green means the policy chases it; red means it avoids it. The win comes from prizing the late-game siege tower and forward pressure, and from refusing to sit on ground it already owns.

Plate II. The learner's win-rate against the four scripted strategies, climbing from nothing over twelve rounds of self-play. The right-hand bars are the traits it learned to seek (green) or to avoid (red). It reached roughly one hundred percent, meaning it found a hole the scripted bots called fair.

IV. READING THE BALANCE

Once the playtesters are strong, the bench runs the match thousands of times and reads the averages. That is **Monte Carlo**, a grand name for a simple habit: do not trust one game, trust ten thousand. The readout below answers the questions that decide whether a game is good. Is the map fair to both sides? Is any single strategy strictly best? Does an early lead simply snowball, or can the loser claw back? On this real terrain the answers are sharp. The map sits close to even, forty-five against fifty-five. But aggression dominates at about three wins in four, far stronger than on flat ground, because the reservoir's chokepoints reward a hard, early push. Pure defence is hopeless, since a turtle can never cross the ravine to score. None of this is a fault to hide. It is a set of dials to turn, and the tool names which to turn and by how much.

BALANCE READOUT

600 matches on the real Lake San Antonio field. Every strategy plays every other, both sides, all seeded.

WIN-RATE BY STRATEGY



FACTION FAIRNESS (IS THE MAP EVEN?)



83% lead holds 17% comeback 11% dead-time 63% sieges fire

MATCH-UP MATRIX

Each cell: the row's win-rate against the column.

	EXP	AGG	BRG	TUR	RL
EXP	54	0	42	100	42
AGG	96	63	100	100	17
BRG	54	4	42	100	29
TUR	0	0	0	0	0
RL	67	46	83	96	0

Plate III. The same ruleset judged over six hundred matches on the Lake San Antonio field. The bars are win-rates by strategy and by side. The grid reads who beats whom, green for the row winning, red for the row losing. The four small

figures below are the texture of a match: how often a lead holds, how often the loser recovers, how much time is dead, and how often a siege actually fires.

V. THE BENCH ITSELF

LEVERS. Ten dials, each written down with what it buys and what it costs (an honesty ledger).

MECHANICS. Capture by presence, five classes, the four-act clock, walls and sieges, fog of war.

PLAYTEST. Thousands of seeded matches, read as averages, never a single lucky game.

BUILT FROM. Plain web code that runs anywhere, real elevation data, every result reproducible.

THE VISION. The Forge will travel from venue to venue, and every field is different. This bench is the pre-flight check. Tune the economy, the classes, and the timing for each new ground in simulation, ship a rulebook that is already close to fair, and spend the few costly real rehearsals on the one thing no computer can judge: whether five hundred living people are having a good time.

THE KIT. Field Manual (the full rules) · live demo at alyasska.github.io/seed-artifact · design and balance notes in the repository at github.com/Alyasska/seed-artifact.